

Title: StegoGAN: A GAN-Based Framework for Steganography Using LSB Embedding

Abstract:

This paper introduces a novel deep learning-based image steganography technique that combines Generative Adversarial Networks (GANs) with the Least Significant Bit (LSB) embedder, used in the CIFAR-10 image dataset for unnoticeable and secure data hiding[1], [2].

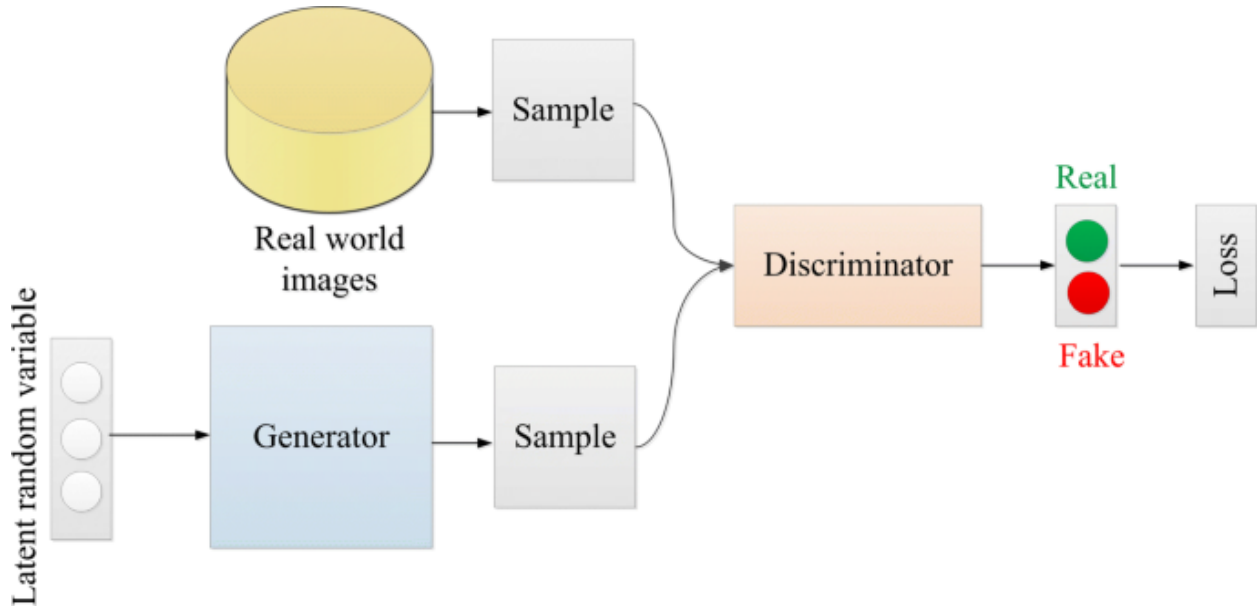


Fig. 1: Structure of generative adversarial network

Steganography, the process of hiding information inside digital media, calls for the balance between concealment performance and image quality. To bypass the limitations of traditional methods, this approach offers a pipeline that initially converts secret textual information into the bitmap image format for pixel-level integration into cover images. Cover images are preprocessed by filtering and edge detection algorithms to identify low perceptual sensitivity areas—perfect areas to embed data. The preprocessing aims to make the embedded information as least distorted as possible so that the image still maintains its original appearance. The GAN model employed here is a generator, which places the bitmap text in the cover image through LSB manipulation, and a discriminator, which learns to distinguish between clean and stego-images. Adversarial training is what propels the generator to produce stego-images increasingly difficult to differentiate from authentic, unchanged images[3], [4], [5], [6], [7], [8], [9].

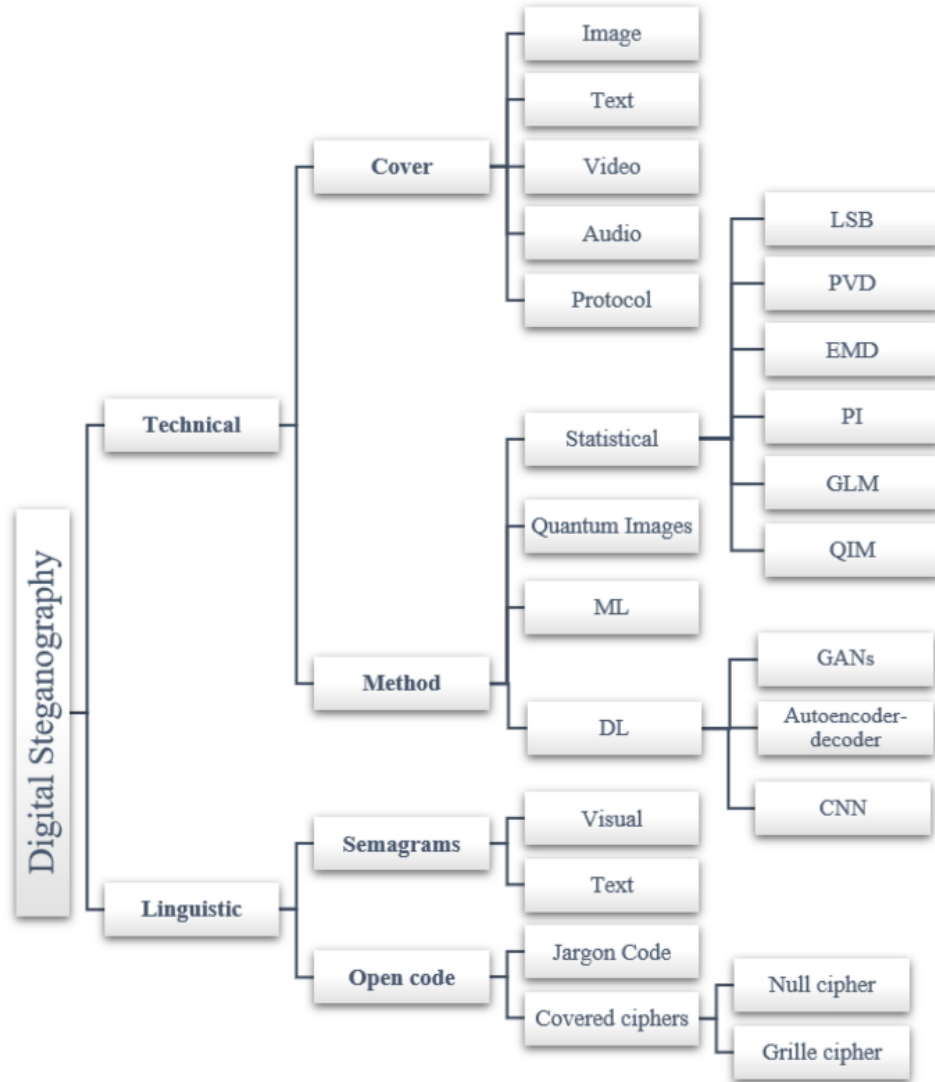


Fig. 2: Taxonomy of steganography

To ensure high-quality output, the model is trained using a combination of loss functions: (1) recovery loss, indicating how well the hidden data can be reconstructed from the embedded data; (2) L2 loss, evaluating pixel-wise mean squared error between original image and stego-image; (3) perceptual loss, evaluating the difference in high-level features extracted by a pre-trained convolutional network, maintaining visual similarity; and (4) adversarial loss, penalizing the generator for producing images detectable as fake by the discriminator. Experimental evaluation on the CIFAR-10 dataset shows that the developed method creates excellent trade-offs between recovery quality and imperceptibility. Edge detection in conjunction with LSB embedding helps mask information in visually less detectable regions, while the GAN's training ensures plausible

output. The system has the ability to produce high PSNR values along with low structural dissimilarity indices, thus keeping visual distortion to a minimum. This paper demonstrates how the integration of classical steganographic approaches with adversarial learning and perceptual metrics provides an effective, safe, and intelligent system of covered communication. This proposed structure is highly universal and constitutes a concrete foundation for viable applications towards enhanced privacy AI, secure messaging, and electronic watermarking[10], [11], [12], [13], [14].

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